Bow Valley College

Course Outline

DATA1203

Acquiring and Preparing Data
Spring 2026 - Current

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Bow Valley College is situated in Treaty 7 territory. We honour the territories and Peoples of: the Blackfoot Confederacy, which includes the Siksika, Kainai, Piikani, and Amskapi Piikani (Blackfeet) First Nations; the Tsuut'ina First Nation; and the Îyâxe Nakoda, which includes the Goodstoney, Chiniki, and Bearspaw First Nations. We also acknowledge the historical Northwest Métis homeland and the Otipemisiwak Métis Government of Alberta, districts 4, 5, and 6 in Battle River territory.

DATA1203 Acquiring and Preparing Data

COURSE DESCRIPTION

This course encompasses the complete workflow required to take raw or external data and transform it into a clean, structured, and usable dataset for analysis. You begin by identifying relevant data sources, evaluating their quality and structure, and applying appropriate extraction methods. You must demonstrate responsible data handling practices, including compliance with ethical and organizational standards, and document any issues that arise during the extraction process. Once acquired, data must be assessed, cleaned, and shaped to meet project goals.You develop and apply strategies for handling common data issues such as inconsistencies, missing values, outliers, and duplicates. You create derived variables, aggregate or merge data as needed, and validate the dataset to ensure it meets analytical requirements. Throughout the process,you are expected to act with diligence, attention to detail, and a commitment to transparency—producing datasets that are well-documented, reproducible, and ready for meaningful analysis in real-world healthcare contexts.

REQUISITES	Complete the following courses: • DATA1100 - Programming for Data Analysis (3)
EQUIVALENTS	None
CREDITS	6
HOURS	90
ELIGIBLE FOR PLAR	No
ZERO TEXTBOOK COST	No

COURSE COMPETENCY

COMPETENCY TITLE

Acquiring and Preparing Data

COMPETENCY STATEMENT

Acquire Data: Identify, evaluate, and extract data from appropriate sources while demonstrating accountability, attention to detail, and ethical data use.

Prepare Data: Prepare datasets for analysis by extracting, structuring, cleaning, enriching, and validating data from local and networked sources using SQL and Python, while demonstrating accuracy,

adaptability, and ethical responsibility.

COMPETENCY DESCRIPTION

Acquire Data: This competency involves locating, assessing, authorizing, and extracting relevant data to support analytical objectives. Learners must demonstrate the ability to identify and evaluate potential data sources based on criteria such as accuracy, completeness, structure, and legal/ethical constraints. The process includes selecting suitable extraction methods (e.g., SQL queries, APIs, web scraping), executing secure and accurate extractions, and documenting any issues encountered. Professionals in data roles are expected to carry out this work with a strong sense of responsibility, data security awareness, and clarity in communication. Whether gathering operational records from a hospital database or pulling public health indicators from an open API, this competency is foundational to any data-driven project — ensuring the right data is available, accessible, and appropriate for analysis.

Prepare Data: This competency focuses on transforming raw or external data into structured, validated, and analysis-ready formats using both SQL and Python. Learners acquire data from local databases or networked APIs, apply filtering and transformation logic, and prepare data in accordance with analytical requirements. They clean datasets by addressing common issues such as outliers, duplicates, missing values, and time-based inconsistencies. Learners are also expected to parse structured data formats like JSON or XML and convert them into tabular forms for analysis. By working with realistic datasets and developing scripts to retrieve and structure data programmatically, learners gain hands-on experience with workplace data preparation workflows. Throughout the process, they demonstrate accuracy, critical thinking, and professional judgment when handling imperfect data, documenting their work, and preparing outputs for use in downstream systems.

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong students. In addition to the discipline-specific skills that students acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others

5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Select data sources	2, 3, 5
2	Extract data	2, 3, 5
3	Determine data preparation requirements	2, 3, 5
4	Restructure data to facilitate analysis	2, 3, 5
5	Clean data to resolve quality issues	2, 3, 5
6	Enrich data for analytical value	2, 3, 5
7	Validate data	2, 3, 5
8	Build a Python program that interacts with APIs	2, 3, 5

LEARNING PATHWAY

**The time it takes learners to demonstrate competencies will vary. An example of a suggested schedule for learning and development is shown below. Learners will need to plan out their assessment attempts within their course. For additional information, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	LEARNING AND DEVELOPMENT PLAN
1	Module 1: Select Data Sources
2	Module 2: Extract Data
3	Module 2: Extract Data
4	Module 2: Extract Data
5	Module 2: Extract Data
6	Module 3: Determine data preparation requirements
7	Module 4: Restructure data to facilitate analysis
8	Module 5: Clean data to resolve quality issues
9	Reading Week
10	Module 5: Clean data to resolve quality issues
11	Module 6: Enrich data for analytical value; Module 7: Validate Data
12	Module 8: Build a Python program that interacts with APIs
13	Complete Competency Assessment
14	Complete Competency Assessment
15	Complete Mastery Assessment

ASSESSMENT

This course follows assessment on demand approach, in which students will be assessed, and receive structured feedback, and a personalized learning plan. Students will also receive differentiated support from an instructor based on their individual needs.

Students will have a variety of ways to demonstrate they have met the required competency through the demonstration of learning outcomes and criteria as laid out in the rubric. Students will have multiple (but not unlimited) attempts to prove competency. It is suggested that students plan out their assessment attempts within their course.

Students will have flexibility in how they satisfy course learning outcomes while still adhering to the criteria found in the rubric and the Course Offering information. Please refer to the Course Offering Information and the rubric in Brightspace for additional information.

COURSE LEARNING OUTCOMES	ASSESSMENT
1, 2, 3, 4, 5, 6, 7, 8	Competency Assessment (8 parts)
1, 2, 3, 4, 5, 6, 7, 8	Mastery Assessment

PERFORMANCE STANDARDS

Successful demonstration of all learning outcomes is required to pass this course. This course may be part of a competency or larger program of study.

Please consult with the program area or the Bow Valley College website for completion requirements.

GRADING SCHEME

Symbol	**Description**	**Grade Point Value**
MAST	Mastery	4.00
COMP	Competent	3.00
DEVL	Developing	N/A
F	Fail	0.00
W	Withdraw	N/A

REQUIRED LEARNING RESOURCES

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Syverson, B., & Murach, J. (2023). Murach's SQL Server 2022 for Developers. Murach Books.

ISBN: 9781943873067 paper copy only

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Students with documented disabilities (e.g., mental health diagnosis, learning disability, physical disability, etc.) may qualify for classroom and/or exam accommodations. The College can also assist students who may be experiencing difficulties with learning but do not have a formal diagnosis. Students who need academic and exam accommodations based on other protected grounds may also make those requests.
It is the student's responsibility to request academic accommodations. Students are encouraged to contact Accessibility Services as early as possible.

Visit: Learner Success Services- 1st floor south campus
Contact: accessibility@bowvalleycollege.ca or 403-410-1440
BVC Central: Accessibility Services

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Students are urged to become familiar with and uphold the following policies: Academic Integrity (500-1-7), Student Code of Conduct (500-1-1), Student Appeals (500-1-12), Attendance (500-1- 10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13).
Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by students is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Academic Integrity:

At Bow Valley College, we value academic integrity. This means everyone at the college, including students, faculty, and staff, follow principles that foster ethical and truthful ways of being in their work and studies. The college is committed to integrity being infused in all aspects of work and studies. It is important to act with integrity in everything you do at the college. The values you practice at the college are the same values that will inform and guide your career. If someone breaks from the principles that are in agreement with academic integrity policy, the college is committed to carefully investigating and fairly handling any issues related to academic integrity and academic misconduct.

Academic misconduct is acting in such a way as to give yourself or others an unfair academic advantage or disadvantage. Examples of this can include:

- Plagiarism
- Cheating on tests or exams
- Tampering with or falsifying any grades or documents
- Sharing assignments or exam information with others
- Aiding others in dishonest acts
- Using artificial intelligence in graded work without the explicit consent of the instructor
- Recording anything in the classroom or other instructional area without permission

Breaking from the principles that guide academic integrity can lead to educative interventions and/or punitive measures. It is the student's responsibility to review the Academic Integrity Policy. Please review the following resources:

- Academic Integrity Policy Summary
- Academic Integrity Policy
- Academic Integrity Procedure

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student work. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where either the course instructor, a representative of Bow Valley College, or a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;

b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, https://www.bowvalleycollege.ca/academic-calendar.
Students are encouraged to keep a copy of this course outline for future reference.

Collection of Personal Information:

This course, including your image and voice, may be recorded and made available to you and other students taking the course. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements. You may use the recordings only for educational purposes. You must not copy, share, or use the recordings for any other purpose without the instructor's express permission. Your personal information is collected in accordance with section 4 (c) of the Protection of Privacy Act, Alberta (POPA) to deliver academic programming, support student flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. Bow Valley College uses automated systems to generate content and make decisions, recommendations and predictions, and your personal information may be included in these automated systems.
Your personal information will be used and disclosed in accordance with sections 12 and 13 of POPA and protected in accordance with section 10 of POPA. If you have any questions about the collection, use, disclosure or protection of personal information, please contact the Access and Privacy Officer at privacy@bowvalleycollege.ca.